

PubH7485 Homework4 Key

2024-10-18

Question 1 (30 points total, 5 for each estimate)

Table 1: ATT estimates using different causal methods

	Weight Change	SBP	DBP
Regression	3.207	1.307	1.334
PS Regression	3.213	1.706	1.299
PSS	3.139	1.973	1.397
IPW	3.238	1.774	1.341
Match 1:1	3.503	2.561	2.047
Match 2:1	2.968	2.34	1.547

The model used for obtaining ATTs could be different (based on the best model you chose from the previous homework), but the estimates should be similar.

Question 2 (20 points total, 10 for plots and 10 for tables)

I created histograms, SMD plot and tableones for diagnosis of the matching process. Other diagnosis (including QQ-plots) are also accepted.

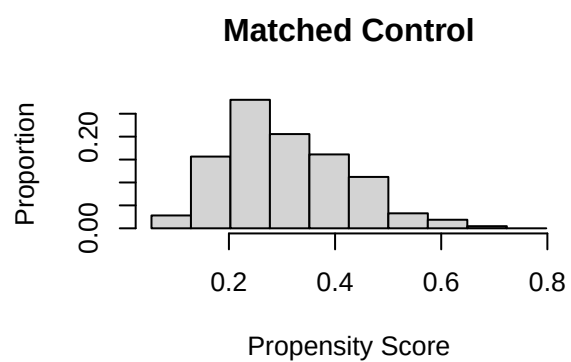
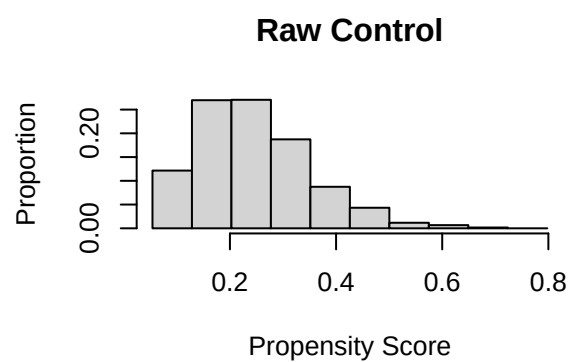
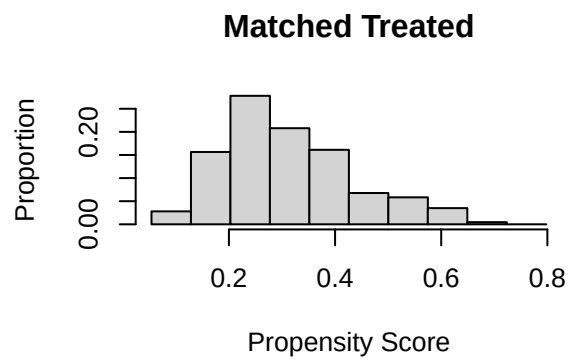
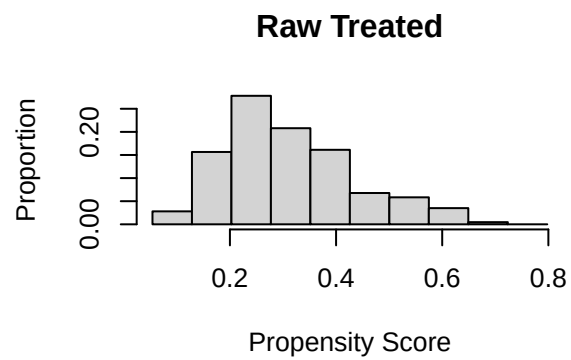
We can see the matching process (for both 1:1 and 2:1 matching) gives us balance of the adjusted covariates between the 'qsmk' groups.

	level	No	Yes	SMD
n		428	428	
sex (%)	Male	250 (58.4)	237 (55.4)	0.061
	Female	178 (41.6)	191 (44.6)	
race (%)	White	392 (91.6)	390 (91.1)	0.017
	Black or Other	36 (8.4)	38 (8.9)	
AGE IN 1971 (mean (SD))		46.32 (12.83)	46.70 (12.52)	0.030
education (%)	8th grade or less	92 (21.5)	93 (21.7)	0.060
	HS dropout	79 (18.5)	78 (18.2)	
	HS	163 (38.1)	164 (38.3)	
	College dropout	36 (8.4)	30 (7.0)	
	College or more	58 (13.6)	63 (14.7)	
NUMBER OF CIGARETTES SMOKED PER DAY IN 1971 (mean (SD))		19.33 (12.11)	18.79 (12.26)	0.044
YEARS OF SMOKING (mean (SD))		26.83 (13.28)	26.61 (13.03)	0.017

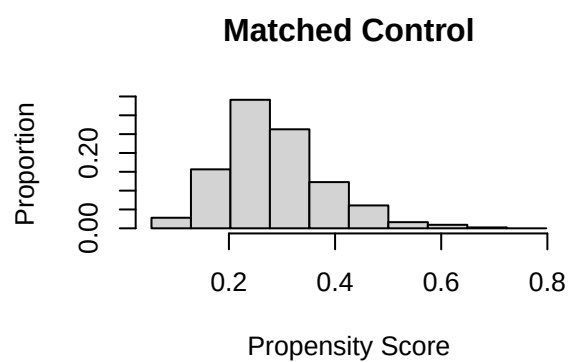
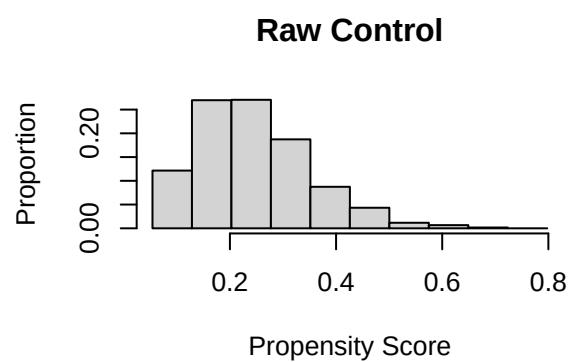
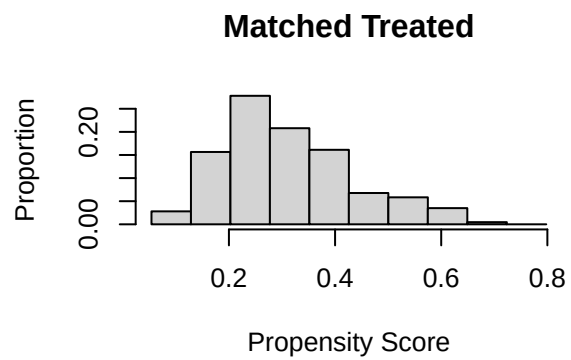
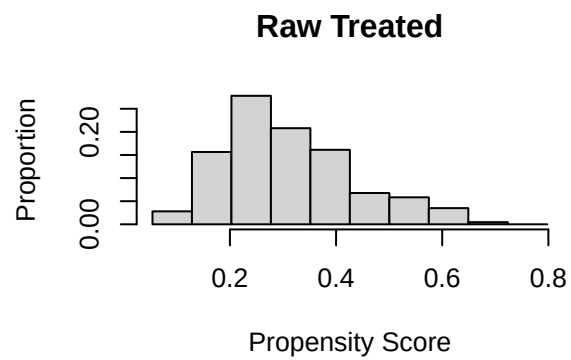
	level	No	Yes	SMD
exercise (%)	Little or no exercise	69 (16.1)	70 (16.4)	0.007
	Moderate exercise	182 (42.5)	181 (42.3)	
	Much exercise	177 (41.4)	177 (41.4)	
active (%)	Very active	193 (45.1)	182 (42.5)	0.061
	Moderately active	185 (43.2)	198 (46.3)	
	Inactive	50 (11.7)	48 (11.2)	
WEIGHT IN KILOGRAMS IN 1971 (mean (SD))		73.47 (16.92)	72.63 (16.08)	0.051
hbp (%)	Never	188 (43.9)	184 (43.0)	0.034
	Ever	38 (8.9)	42 (9.8)	
	Missing	202 (47.2)	202 (47.2)	

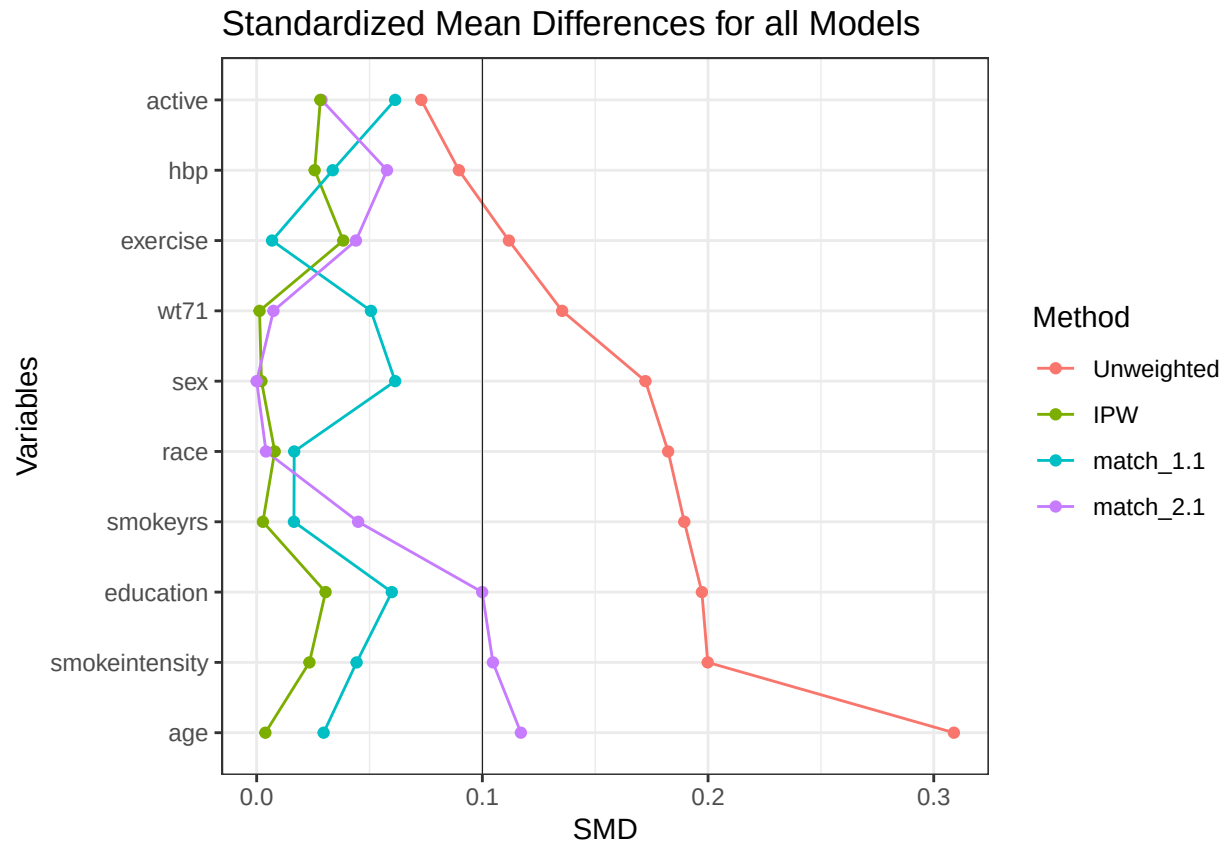
	level	No	Yes	SMD
n		856	428	
sex (%)	Male	474 (55.4)	237 (55.4)	<0.001
	Female	382 (44.6)	191 (44.6)	
race (%)	White	781 (91.2)	390 (91.1)	0.004
	Black or Other	75 (8.8)	38 (8.9)	
AGE IN 1971 (mean (SD))		45.25 (12.11)	46.70 (12.52)	0.117
education (%)	8th grade or less	164 (19.2)	93 (21.7)	0.100
	HS dropout	168 (19.6)	78 (18.2)	
	HS	340 (39.7)	164 (38.3)	
	College dropout	73 (8.5)	30 (7.0)	
	College or more	111 (13.0)	63 (14.7)	
NUMBER OF CIGARETTES SMOKED PER DAY IN 1971 (mean (SD))		20.06 (11.87)	18.79 (12.26)	0.105
YEARS OF SMOKING (mean (SD))		26.04 (12.32)	26.61 (13.03)	0.045
exercise (%)	Little or no exercise	148 (17.3)	70 (16.4)	0.044
	Moderate exercise	372 (43.5)	181 (42.3)	
	Much exercise	336 (39.3)	177 (41.4)	
active (%)	Very active	372 (43.5)	182 (42.5)	0.029
	Moderately active	395 (46.1)	198 (46.3)	
	Inactive	89 (10.4)	48 (11.2)	
WEIGHT IN KILOGRAMS IN 1971 (mean (SD))		72.51 (15.71)	72.63 (16.08)	0.007
hbp (%)	Never	378 (44.2)	184 (43.0)	0.058
	Ever	70 (8.2)	42 (9.8)	
	Missing	408 (47.7)	202 (47.2)	

[1] "For 1:1 Matching"



```
## [1] "For 2:1 Matching"
```





Appendix - R code

Question 1

1. Load packages and impute the dataset

```
set.seed(7485)

library(dplyr)
library(labelled)
library(causaldata)
library(knitr)
library(MASS)
library(reshape2)
library(ggplot2)
library(survey)
library(tableone)

library(formattable)
library(rms)
library(forestplot)
library(Matching)
library(MatchIt)
```

```

data('nhefs')

# treatment
nhefs$qsmk
# outcomes of interest
nhefs$wt82_71
nhefs$sbp
nhefs$dbp

all_covars <- setdiff(names(nhefs), c('dadth', 'death', 'hightax82', 'modth', 'price71_82',
    'price82', 'seqn', 'yrdth', 'wt82', 'tax71_82',
    'tax82', 'smokeyrs', 'smkintensity82_71',
    'birthplace', 'id', 'censored', 'older'))

adj_vars <- c('sex', 'race', 'age', 'education', 'smokeintensity', 'smokeyrs', 'exercise', 'active', 'w

# Convert categorical variables to factors and set levels for them in nhefs data frame
nhefs$active <- factor(nhefs$active, levels = c(0, 1, 2), labels = c("Very active", "Moderately active", "Little or no exercise"))
nhefs$alcoholfreq <- factor(nhefs$alcoholfreq, levels = c(0, 1, 2, 3, 4, 5), labels = c("Almost every day", "Several times a week", "Once a week", "A few times a month", "Once a month or less often", "Never"))
nhefs$alcoholpoly <- factor(nhefs$alcoholpoly, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$alcoholtype <- factor(nhefs$alcoholtype, levels = c(1, 2, 3, 4), labels = c("Beer", "Wine", "Liquor", "Other"))
nhefs$allergies <- factor(nhefs$allergies, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$asthma <- factor(nhefs$asthma, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$birthcontrol <- factor(nhefs$birthcontrol, levels = c(0, 1, 2), labels = c("No", "Yes", "Missing"))
nhefs$boweltrouble <- factor(nhefs$boweltrouble, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$bronch <- factor(nhefs$bronch, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$chroniccough <- factor(nhefs$chroniccough, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$colitis <- factor(nhefs$colitis, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$diabetes <- factor(nhefs$diabetes, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$education <- factor(nhefs$education, levels = c(1, 2, 3, 4, 5), labels = c("8th grade or less", "9th-11th grade", "12th grade", "Some college", "Bachelor's degree or higher"))
nhefs$exercise <- factor(nhefs$exercise, levels = c(0, 1, 2), labels = c("Little or no exercise", "Moderate exercise", "Vigorous exercise"))
nhefs$hayfever <- factor(nhefs$hayfever, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$hbp <- factor(nhefs$hbp, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$hbpmed <- factor(nhefs$hbpmed, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$headache <- factor(nhefs$headache, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$hepatitis <- factor(nhefs$hepatitis, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$hf <- factor(nhefs$hf, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$marital <- factor(nhefs$marital, levels = c(1, 2, 3, 4, 5, 6, 8), labels = c("Under 17", "Married", "Widowed", "Divorced", "Separated", "Never married", "Other"))
nhefs$nerves <- factor(nhefs$nerves, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$nervousbreak <- factor(nhefs$nervousbreak, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$otherpain <- factor(nhefs$otherpain, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$pepticulcer <- factor(nhefs$pepticulcer, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$pica <- factor(nhefs$pica, levels = c(0, 1, 2), labels = c("Never", "Ever", "Missing"))
nhefs$polio <- factor(nhefs$polio, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$qsmk <- factor(nhefs$qsmk, levels = c(0, 1), labels = c("No", "Yes"))
nhefs$race <- factor(nhefs$race, levels = c(0, 1), labels = c("White", "Black or Other"))
nhefs$sex <- factor(nhefs$sex, levels = c(0, 1), labels = c("Male", "Female"))
nhefs$tb <- factor(nhefs$tb, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$tumor <- factor(nhefs$tumor, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$weakheart <- factor(nhefs$weakheart, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$wtloss <- factor(nhefs$wtloss, levels = c(0, 1), labels = c("Never", "Ever"))
nhefs$lackpep <- factor(nhefs$lackpep, levels = c(0, 1), labels = c("Never", "Ever"))

```

```

nhefs$infection <- factor(nhefs$infection, levels = c(0, 1), labels = c("Never", "Ever"))

mode_function <- function(x) {
  x <- as.character(x)
  tbl <- table(x, useNA = "ifany")
  as.character(names(tbl)[which.max(tbl)])
}

for (var in all_covars) {
  if (is.numeric(nhefs[[var]])) {
    # Impute missing values for numeric variables with the median
    nehs[[var]][is.na(nhefs[[var]])] <- median(nhefs[[var]], na.rm = TRUE)
    print(paste("Imputed numeric variable:", var))
  } else if (is.factor(nhefs[[var]]) || is.character(nhefs[[var]])) {
    # Impute missing values for categorical variables with the mode
    nehs[[var]][is.na(nhefs[[var]])] <- mode_function(nhefs[[var]])
    # nehs[[var]] <- as.factor(nhefs[[var]]) # Ensure it's still a factor
    print(paste("Imputed categorical variable:", var))
  }
}

```

2. Obtain ATT estimates

```

m1 <- glm(wt82_71 ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp), data = nehs)

m2 <- glm(sbp ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp), data = nehs)

m3 <- glm(dbp ~ qsmk * (sex + race + age + education + smokeintensity + smokeyrs +
  exercise + active + wt71 + hbp), data = nehs)

p1 <- glm(qsmk ~ sex + race + poly(age, 2) + education + poly(smokeintensity, 2) + poly(smokeyrs, 2) +
  exercise + active + poly(wt71, 2) + hbp, family = binomial(), data = nehs)

# Regression
data_trt <- data_ctr <- nehs[nhefs$qsmk == 'Yes', ]

data_trt$qsmk = 'Yes'; data_trt$qsmk <- as.factor(data_trt$qsmk)
data_ctr$qsmk = 'No'; data_ctr$qsmk <- as.factor(data_ctr$qsmk)

pred1_wc <- predict(m1, newdata = data_trt, type = "response")
pred0_wc <- predict(m1, newdata = data_ctr, type = "response")
(ATT_wc <- mean(pred1_wc - pred0_wc))

pred1_sbp <- predict(m2, newdata = data_trt, type = "response")
pred0_sbp <- predict(m2, newdata = data_ctr, type = "response")
(ATT_sbp <- mean(pred1_sbp - pred0_sbp))

pred1_dbp <- predict(m3, newdata = data_trt, type = "response")
pred0_dbp <- predict(m3, newdata = data_ctr, type = "response")

```

```

(ATT_dbp <- mean(pred1_dbp - pred0_dbp))

# PS regression
nhefs$ps <- predict(p1, type = "response")

m1.ps <- glm(wt82_71 ~ qsmk*rccs(ps, 5), data = nhefs)
m2.ps <- glm(sbp ~ qsmk*rccs(ps, 5), data = nhefs)
m3.ps <- glm(dbp ~ qsmk*rccs(ps, 5), data = nhefs)

data_trt_alt <- data_ctr_alt <- nhefs
data_trt_alt$qsmk = 'Yes'
data_ctr_alt$qsmk = 'No'

pred1.ps_wc <- predict(m1.ps, newdata = data_trt_alt, type = "response")
pred0.ps_wc <- predict(m1.ps, newdata = data_ctr_alt, type = "response")
(ATT.ps_wc <- mean(pred1.ps_wc[nhefs$qsmk == 'Yes'] - pred0.ps_wc[nhefs$qsmk == 'Yes']))

pred1.ps_sbp <- predict(m2.ps, newdata = data_trt_alt, type = "response")
pred0.ps_sbp <- predict(m2.ps, newdata = data_ctr_alt, type = "response")
ATT.ps_sbp <- mean(pred1.ps_sbp[nhefs$qsmk == 'Yes'] - pred0.ps_sbp[nhefs$qsmk == 'Yes'])

pred1.ps_dbp <- predict(m3.ps, newdata = data_trt_alt, type = "response")
pred0.ps_dbp <- predict(m3.ps, newdata = data_ctr_alt, type = "response")
ATT.ps_dbp <- mean(pred1.ps_dbp[nhefs$qsmk == 'Yes'] - pred0.ps_dbp[nhefs$qsmk == 'Yes'])

# PS stratification
ps <- predict(p1, type = "response")

ps_quintile <- cut(ps,
  breaks = c(0, quantile(ps[nhefs$qsmk == 'Yes'], p = c(0.2, 0.4, 0.6, 0.8)), 1), labels = 1:5)
nA <- nrow(nhefs[nhefs$qsmk == 'Yes', ])
nAj <- table(ps_quintile[nhefs$qsmk == 'Yes'])

te_quintile_wc <- tapply(nhefs$wt82_71[nhefs$qsmk == 'Yes'], ps_quintile[nhefs$qsmk == 'Yes'], mean) -
  tapply(nhefs$wt82_71[nhefs$qsmk == 'No'], ps_quintile[nhefs$qsmk == 'No'], mean)
(ATT_PSS_wc <- sum(te_quintile_wc * nAj / nA))

te_quintile_sbp <- tapply(nhefs$sbp[nhefs$qsmk == 'Yes'], ps_quintile[nhefs$qsmk == 'Yes'], mean) -
  tapply(nhefs$sbp[nhefs$qsmk == 'No'], ps_quintile[nhefs$qsmk == 'No'], mean)
(ATT_PSS_sbp <- sum(te_quintile_sbp * nAj / nA))

te_quintile_dbp <- tapply(nhefs$dbp[nhefs$qsmk == 'Yes'], ps_quintile[nhefs$qsmk == 'Yes'], mean) -
  tapply(nhefs$dbp[nhefs$qsmk == 'No'], ps_quintile[nhefs$qsmk == 'No'], mean)
(ATT_PSS_dbp <- sum(te_quintile_dbp * nAj / nA))

# IPW
w1 <- as.numeric(nhefs$qsmk)-1
w0 <- (1-(as.numeric(nhefs$qsmk)-1))/(1-ps)*ps
(ATT_IPW_wc <- weighted.mean(nhefs$wt82_71, w = w1) - weighted.mean(nhefs$wt82_71, w=w0))
(ATT_IPW_sbp <- weighted.mean(nhefs$sbp, w = w1) - weighted.mean(nhefs$sbp, w=w0))
(ATT_IPW_dbp <- weighted.mean(nhefs$dbp, w = w1) - weighted.mean(nhefs$dbp, w=w0))

```



```

## matching
# 1:1
rownames(nhefs) <- 1:nrow(nhefs)
mod_match1 <- matchit(qsmk ~ sex + race + poly(age, 2) + education + poly(smokeintensity, 2) +
                      poly(smokeyrs, 2) + exercise + active + poly(wt71, 2) + hbp,
                      distance = "logit", method = "nearest", data = neefs, ratio = 1)
print(tail(mod_match1$match.matrix))

nhefs_match <- data.frame(rbind(nhefs[nhefs$qsmk == 'Yes', ], neefs[as.numeric(mod_match1$match.matrix)
tabmatched1.1 <- CreateTableOne(vars = adj_vars, strata = "qsmk",
                                data = neefs_match, test = FALSE)
t1.1 <- print(tabmatched1.1, smd = TRUE, showAllLevels = TRUE, varLabels = TRUE)

# plot(mod_match1, type = "hist")
# plot(mod_match1, type = "QQ")

n1 <- table(nhefs$qsmk)[2]
n0 <- length(as.numeric(mod_match1$match.matrix))

p1_wc <- mean(nhefs$wt82_71[nhefs$qsmk == 'Yes'])
p0_wc <- mean(nhefs$wt82_71[as.numeric(mod_match1$match.matrix)])
p1_sbp <- mean(nhefs$sbp[nhefs$qsmk == 'Yes'])
p0_sbp <- mean(nhefs$sbp[as.numeric(mod_match1$match.matrix)])
p1_dbp <- mean(nhefs$dbp[nhefs$qsmk == 'Yes'])
p0_dbp <- mean(nhefs$dbp[as.numeric(mod_match1$match.matrix)])

(ATT1match_wc <- p1_wc - p0_wc)
(ATT1match_sbp <- p1_sbp - p0_sbp)
(ATT1match_dbp <- p1_dbp - p0_dbp)

# 2:1
mod_match2 <- matchit(qsmk ~ sex + race + poly(age, 2) + education + poly(smokeintensity, 2) +
                      poly(smokeyrs, 2) + exercise + active + poly(wt71, 2) + hbp,
                      distance = "logit", method = "nearest", data = neefs, ratio = 2)
print(tail(mod_match2$match.matrix))

nhefs_match2 <- data.frame(rbind(nhefs[nhefs$qsmk == 'Yes', ], neefs[as.numeric(mod_match2$match.matrix)
tabmatched2.1 <- CreateTableOne(vars = adj_vars, strata = "qsmk",
                                data = neefs_match2, test = FALSE)
t2.1 <- print(tabmatched2.1, smd = TRUE, showAllLevels = TRUE, varLabels = TRUE)

# plot(mod_match2, type = "hist")
# plot(mod_match2, type = "QQ")

n1 <- table(nhefs$qsmk)[2]
n0_2.1 <- length(as.numeric(mod_match2$match.matrix))

p1_wc_2.1 <- mean(nhefs$wt82_71[nhefs$qsmk == 'Yes'])
p0_wc_2.1 <- mean(nhefs$wt82_71[as.numeric(mod_match2$match.matrix)])
p1_sbp_2.1 <- mean(nhefs$sbp[nhefs$qsmk == 'Yes'])
p0_sbp_2.1 <- mean(nhefs$sbp[as.numeric(mod_match2$match.matrix)])
p1_dbp_2.1 <- mean(nhefs$dbp[nhefs$qsmk == 'Yes'])
p0_dbp_2.1 <- mean(nhefs$dbp[as.numeric(mod_match2$match.matrix)])

```

```

(ATT2match_wc <- p1_wc_2.1 - p0_wc_2.1)
(ATT2match_sbp <- p1_sbp_2.1 - p0_sbp_2.1)
(ATT2match_dbp <- p1_dbp_2.1 - p0_dbp_2.1)

ATT_reg <- c(ATT_wc, ATT_sbp, ATT_dbp)
ATT_ps <- c(ATT.ps_wc, ATT.ps_sbp, ATT.ps_dbp)
ATT_pss <- c(ATT_PSS_wc, ATT_PSS_sbp, ATT_PSS_dbp)
ATT_ipw <- c(ATT_IPW_wc, ATT_IPW_sbp, ATT_IPW_dbp)
ATT_match_11 <- c(ATT1match_wc, ATT1match_sbp, ATT1match_dbp)
ATT_match_21 <- c(ATT2match_wc, ATT2match_sbp, ATT2match_dbp)

tabletext <- rbind(c("", "Weight Change", "SBP", "DBP"),
  c("Regression", round(ATT_reg, digits = 3)),
  c("PS Regression", round(ATT_ps, digits = 3)),
  c("PSS", round(ATT_pss, digits=3)),
  c("IPW", round(ATT_ipw, digits=3)),
  c("Match 1:1", round(ATT_match_11, digits=3)),
  c("Match 2:1", round(ATT_match_21, digits=3))
)

kable(as.data.frame(tabletext))

```

Question 2

1. Table and plot diagnosis for the matching process

```

# table diagnosis
kable(t1.1)
kable(t2.1)

# plot diagnosis
plot(mod_match1, type = "hist")
plot(mod_match2, type = "hist")

# SMD plot
nhfs$weight_a <- with(nhfs, ifelse(qsmk == "Yes", 1 / ps, 1 / (1 - ps)))
dataSvy_a <- svydesign(ids = ~ 1, data = nhfs, weights = ~ weight_a)
tabUnweighted_a <- CreateTableOne(vars = adj_vars, strata = "qsmk", data = nhfs, test = FALSE)
tabWeighted_a <- svyCreateTableOne(vars = adj_vars, strata = "qsmk", data = dataSvy_a, test = FALSE)

dataPlot_a <- data.frame(variable = rownames(ExtractSmd(tabUnweighted_a)),
  Unweighted = as.numeric(ExtractSmd(tabUnweighted_a)),
  IPW = as.numeric(ExtractSmd(tabWeighted_a)))

dataPlotMelt_a <- melt(data = dataPlot_a,
  id.vars = c("variable"),
  variable.name = "Method",
  value.name = "SMD")

dataPlot_b <- data.frame(variable = rownames(ExtractSmd(tabmatched1.1)),

```

```

        Unweighted = as.numeric(ExtractSmd(tabmatched1.1)),
        match_1.1 = as.numeric(ExtractSmd(tabmatched1.1)))

dataPlotMelt_b <- melt(data = dataPlot_b,
                      id.vars = c("variable"),
                      variable.name = "Method",
                      value.name = "SMD")

dataPlot_c <- data.frame(variable = rownames(ExtractSmd(tabmatched2.1)),
                        Unweighted = as.numeric(ExtractSmd(tabmatched2.1)),
                        match_2.1 = as.numeric(ExtractSmd(tabmatched2.1)))

dataPlotMelt_c <- melt(data = dataPlot_c,
                      id.vars = c("variable"),
                      variable.name = "Method",
                      value.name = "SMD")

dataPlotMelt <- rbind(dataPlotMelt_a, dataPlotMelt_b[11:20, ], dataPlotMelt_c[11:20, ])
dataPlotMelt %>%
  arrange(desc(SMD), variable) %>%
  mutate(variable_ordered = factor(variable, levels = unique(variable))) %>%
  ggplot(aes(x = variable_ordered, y = SMD, group = Method, color = Method)) +
  geom_line() +
  geom_point() +
  geom_hline(yintercept = 0.1, color = "black", linewidth = 0.1) +
  coord_flip() +
  theme_bw() +
  theme(legend.key = element_blank()) + xlab('Variables') +
  labs(title="Standardized Mean Differences for all Models")

```